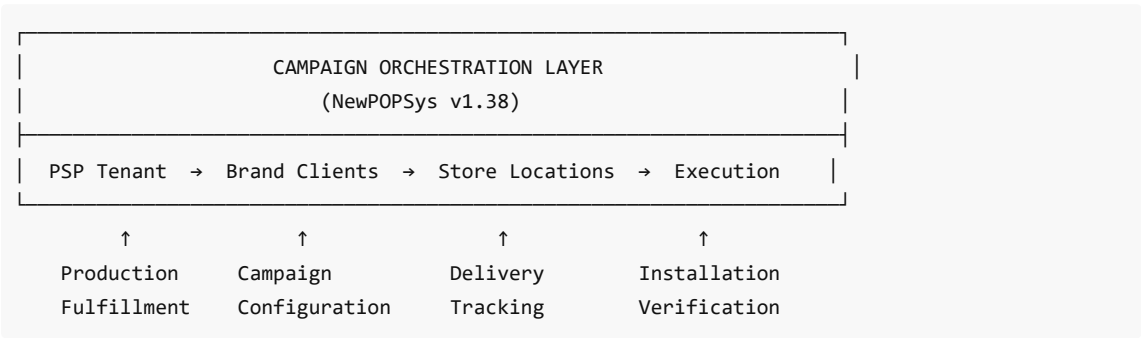


2.1 Product Perspective

System Context

NewPOPSys v1.38 is a standalone multi-tenant SaaS platform designed to replace spreadsheet-and-email workflows for POP (Point-of-Purchase) campaign management. The system positions Print Service Providers (PSPs) as the orchestrating authority for campaign lifecycle management across multiple Brands and their retail Stores.

Market Position



System Boundaries

NewPOPSys operates as the coordination layer between:

External System	Integration Direction	Interface Type
PSP MIS/ERP Systems	Bidirectional	Webhooks + API
Brand Client Systems	Inbound	API + CSV Import
Shipping Carriers	Outbound	Tracking Data Import
Cloud Storage (S3)	Bidirectional	Direct Upload/Download
Email Services	Outbound	SMTP/SES

External Interfaces

User Interfaces

Interface	Technology	Primary Users
Web Application	Next.js (React)	PSP Admins, Brand Admins, Regional Managers
Mobile PWA	Next.js PWA	Store Managers, Store Operators
Admin Console	Next.js	Platform Admins

Hardware Interfaces

NewPOPSys is a cloud-native application with no direct hardware interfaces. However, the system interacts with:

Device Type	Interaction	Requirements
Mobile Devices	Photo capture, offline execution	Camera, GPS (optional), modern browser
Barcode Scanners	Kit item verification	HID-compatible USB/Bluetooth
Label Printers	Shipping label generation	ZPL-compatible (via PSP MIS)

Software Interfaces

External System	Protocol	Purpose
PostgreSQL 15+	SQL/TCP	Primary data store
Redis 7+	Redis Protocol	Session cache, job queue
AWS S3	REST/SDK	Asset storage (photos, exports)
AWS CloudFront	HTTPS	CDN for static assets
AWS SES	SMTP/API	Transactional email
OpenTelemetry	OTLP	Observability
Sentry	HTTPS	Error tracking

Communication Interfaces

Protocol	Usage	Security
HTTPS/TLS 1.3	All external traffic	Required
WebSocket	Real-time notifications	WSS only
REST API	System integration	JWT + API Key
Webhooks	Event dispatch	HMAC-SHA256 signing

Product Position Statement

For Print Service Providers managing POP campaigns for retail brands,
Who need to coordinate campaign execution across hundreds of store locations,
NewPOPSys is a campaign orchestration platform
That provides deterministic lifecycle management with complete audit visibility,
Unlike spreadsheet-and-email coordination or generic project management tools,
Our product enforces structured workflows, captures proof of execution, and maintains compliance traceability.

Relationship to Other Systems

What NewPOPSys IS NOT

Category	Explicit Exclusion	Rationale
MIS/ERP	No job costing, production scheduling, or accounting	Integration point, not replacement
Marketplace	No installer bidding, vendor matching, or ratings	Single PSP model only
Analytics Platform	No predictive analytics or AI insights	Data readiness for future phases
Data Warehouse	No archival storage beyond 90 days	Export pathways provided
Store Customization	Limited store UX flexibility	PSP adoption prioritized

System Dependencies

```
graph TD
  subgraph "NewPOPSys Platform"
    A[Web Application]
    B[API Layer]
    C[Background Jobs]
  end

  subgraph "AWS Infrastructure"
    D[(RDS PostgreSQL)]
    E[(ElastiCache Redis)]
    F[S3 Storage]
    G[CloudFront CDN]
    H[SES Email]
  end

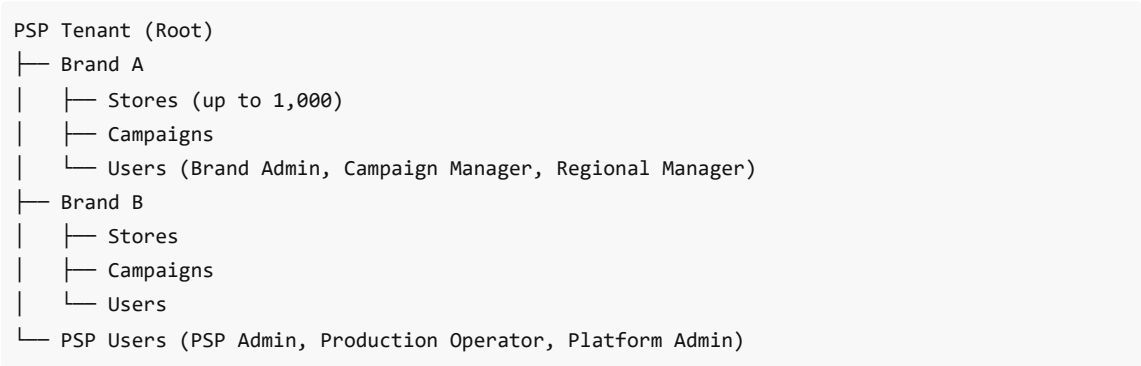
  subgraph "External Systems"
    I[PSP MIS]
    J[Shipping Carriers]
    K[Brand Systems]
  end

  A --> B
  B --> D
  B --> E
  B --> F
  C --> D
  C --> E
  C --> F
  C --> H

  B <--> I
  B <--> K
  J --> B
  F --> G --> A
```

Operating Model

Multi-Tenant Architecture



Data Isolation

Level	Isolation Mechanism	Scope
Tenant	psp_tenant_id column	All data tables
Brand	brand_id column	Brand-scoped tables
Store	store_id column	Store-scoped tables
User	RBAC + Row-level policies	All access

Constraints from External Systems

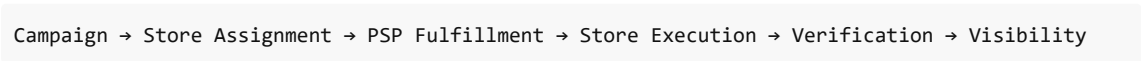
System	Constraint	Impact
PSP MIS	Async integration only	Eventual consistency for order status
Shipping Carriers	Tracking data format varies	Normalization layer required
Mobile Devices	Offline-capable requirement	Sync-on-open architecture
Browser Support	Modern browsers only	No IE11 support

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2.2 Product Functions

Core Platform Loop

NewPOPSys v1.38 implements a deterministic campaign lifecycle with complete audit visibility:



This loop replaces ad-hoc spreadsheet coordination with structured, traceable workflows.

Functional Modules Summary

Module 1: Identity & RBAC

Function	Description
F-1.1 Multi-Tenant Management	PSP as root tenant with Brand hierarchy
F-1.2 User Authentication	Email/password login with secure sessions
F-1.3 Role-Based Access Control	9 personas with scoped permissions
F-1.4 User Invitations	Email-based onboarding workflow
F-1.5 API Key Management	Service account credentials for integrations
F-1.6 Session Security	JWT tokens with refresh mechanism

SUPP Reference: SUPP-003 (RBAC and Permissions Matrix)

Module 2: Stores, Regions & Groups

Function	Description
F-2.1 Store Profile Management	CRUD for store locations (1,000+ per brand)
F-2.2 Geographic Hierarchy	Region > District > Territory structure
F-2.3 Custom Store Grouping	Include/exclude logic for targeting
F-2.4 Store Layout Definition	Physical location slots for POP placement
F-2.5 External ID Aliasing	Brand system ID mapping

SUPP Reference: SUPP-013 (Stores, Regions, Groups)

Module 3: Survey Builder & Photo Rules

Function	Description
F-3.1 Survey Template Creation	Dynamic field configuration
F-3.2 Survey Versioning	Immutable snapshots for campaign pinning
F-3.3 Photo Rule Configuration	Min/max photos, resolution, instructions
F-3.4 Reference Image Management	Example photos for guidance
F-3.5 Conditional Logic	Show/hide based on responses
F-3.6 Repeatable Sections	Per-item or per-slot repetition

SUPP Reference: SUPP-014 (Survey Builder, Layout, Photo Rules), SUPP-037 (Survey Builder)

Module 4: Campaigns, Kits & Assignment

Function	Description
F-4.1 Campaign Creation	Promotional (expiring) and Core Branding types
F-4.2 Kit Definition	Item specifications with SKUs
F-4.3 Store Selection Recipes	Rule-based store targeting
F-4.4 Assignment Pinning	Lock survey/layout versions per store
F-4.5 Campaign Publishing	Transition to active, generate orders
F-4.6 Lifecycle Management	Draft → Active → Complete → Expired

SUPP Reference: SUPP-015 (Campaigns, Kits, Assignment)

Module 5: Orders, Shipments & PSP Operations

Function	Description
F-5.1 Order Generation	Deterministic creation on campaign publish
F-5.2 Partial Shipment Support	Multiple shipments per order
F-5.3 Batch Tracking	PRODUCTION, PICK_PACK, SHIP_WAVE, CUSTOM
F-5.4 Multi-Tracking Numbers	Multiple carriers per shipment
F-5.5 PSP Dashboard	Production queue and status overview
F-5.6 MIS Integration Points	Webhook dispatch for external systems

SUPP Reference: SUPP-016 (Orders, Shipments, Batches)

Module 6: Store Execution & Proof Capture

Function	Description
F-6.1 Mobile PWA Interface	Touch-optimized store execution
F-6.2 Offline Draft Storage	Best-effort offline with sync-on-open
F-6.3 Receive/Verify Workflow	Shipment receipt confirmation
F-6.4 Pre-Install Checklist	Site preparation verification
F-6.5 Item Installation Tracking	Per-item completion status
F-6.6 Proof Photo Capture	Camera integration per item/slot
F-6.7 Completion Attestation	User acknowledgment checkbox

SUPP Reference: SUPP-017 (Store Execution, Proof Capture), SUPP-011 (Offline and Sync)

Module 7: Verification & Retake Loop

Function	Description
----------	-------------

F-7.1 Photo Review Queue	Brand/Regional manager interface
F-7.2 Approval/Rejection	Decision with reason codes
F-7.3 Retake Request Workflow	Notification and re-submission
F-7.4 Verification Modes	STRICT (explicit) vs FAST (auto-approve)
F-7.5 Slot-Level Tracking	Granular verification status

SUPP Reference: SUPP-018 (Verification, Photo Review, Retake)

Module 8: Issues, Reorders & Deinstall

Function	Description
F-8.1 Issue Reporting	Missing, Damaged, Incorrect, Packaging
F-8.2 Issue Triage	Admin review and approval
F-8.3 Reorder Generation	Replacement item ordering
F-8.4 Reorder Tracking	Separate fulfillment lifecycle
F-8.5 Campaign Expiration	End-date handling for promotional
F-8.6 Deinstall Tasks	Removal workflow for expired materials

SUPP Reference: SUPP-019 (Issues, Reorders, Expiration, Deinstall)

Module 9: Exports & Reports

Function	Description
F-9.1 CSV/XLSX Export	Tabular data export
F-9.2 PDF Report Generation	Formatted compliance reports
F-9.3 JSON/XML Data Export	Structured data interchange
F-9.4 Async Processing	BullMQ-based job queue
F-9.5 Export Artifact Storage	S3-based file management
F-9.6 Export History	Download tracking and expiration

SUPP Reference: SUPP-005 (Exports, Reports, Output Contracts)

Module 10: Integrations & Webhooks

Function	Description
F-10.1 Webhook Dispatch	Signed event notifications
F-10.2 Inbound API	Idempotent request handling
F-10.3 Event Outbox Pattern	Reliable event delivery

F-10.4 Retry Scheduling	Exponential backoff with dead-letter
F-10.5 PSP MIS Scaffolding	Integration templates

SUPP Reference: SUPP-006 (Webhooks and Inbound API)

Module 11: Data Retention

Function	Description
F-11.1 90-Day Retention	Post campaign completion window
F-11.2 Export-Before-Delete	Warning and export workflow
F-11.3 Automated Purge Jobs	Scheduled cleanup processing
F-11.4 Asset Registry	Permanent definition preservation
F-11.5 Audit Event Preservation	Compliance trail retention

SUPP Reference: SUPP-008 (Data Retention Classification)

Module 12: Offline & Sync Strategy

Function	Description
F-12.1 PWA Service Worker	Background sync capability
F-12.2 Offline Draft Storage	Local persistence for in-progress work
F-12.3 Sync-on-Open Queue	Automatic submission on reconnect
F-12.4 Conflict Resolution	Last-write-wins with user notification

SUPP Reference: SUPP-011 (Offline and Sync Strategy)

Function Priority Matrix

Priority	Functions	Rationale
P0 - Critical Path	F-1.x, F-4.1-4.5, F-5.1-5.2, F-6.1-6.7, F-7.1-7.3	Core campaign loop
P1 - Essential	F-2.x, F-3.x, F-8.1-8.4, F-10.1-10.4	Complete workflow
P2 - Important	F-9.x, F-11.x, F-8.5-8.6	Operational completeness
P3 - Scaffold Only	F-12.x, Advanced F-10.5	Future enhancement readiness

Cross-Functional Workflows

Campaign Lifecycle

```
stateDiagram-v2
    [*] --> Draft
```

```
Draft --> Scheduled : Publish
Scheduled --> Active : Start Date
Active --> PendingCompletion : All Stores Complete
PendingCompletion --> Complete : Verification Done
Complete --> Expired : End Date (Promotional)
Complete --> [*] : Core Branding
Expired --> [*]
```

Store Assignment Lifecycle

```
stateDiagram-v2
    [*] --> Assigned
    Assigned --> InTransit : Shipped
    InTransit --> Delivered : Received
    Delivered --> InProgress : Started
    InProgress --> Submitted : Completion
    Submitted --> Approved : Verification Pass
    Submitted --> NeedsRetake : Rejection
    NeedsRetake --> Submitted : Resubmit
    Approved --> [*]
```

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2.3 User Classes and Characteristics

User Class Hierarchy

NewPOPSys v1.38 implements a three-tier user hierarchy aligned with the multi-tenant architecture:

```
Platform Level (System-wide)
├─ Platform Admin
|
PSP Level (Tenant-wide)
├─ PSP Admin
├─ Production Operator
|
Brand Level (Brand-scoped)
├─ Brand Admin
├─ Campaign Manager
├─ Regional Manager
|
Store Level (Store-scoped)
├─ Store Manager
├─ Store Operator
|
System Level (Machine-to-Machine)
├─ Integration User
```

User Class Summary

Persona	Level	Access Scope	Primary Interface	Frequency
Platform Admin	Platform	All tenants, all data	Admin Console	Daily
PSP Admin	PSP	Tenant-wide	Web Application	Daily
Production Operator	PSP	Orders, Shipments, Batches	Web Application	Hourly
Brand Admin	Brand	Assigned brands	Web Application	Daily
Campaign Manager	Brand	Assigned campaigns	Web Application	Daily
Regional Manager	Brand	Assigned regions	Web Application	Daily
Store Manager	Store	Assigned stores	Mobile PWA	Per campaign
Store Operator	Store	Assigned stores	Mobile PWA	Per campaign
Integration User	System	API access only	REST API	Automated

Detailed User Classes

Platform Admin

Attribute	Description
Description	Highest privilege system administrator with full access
Population	1-3 per deployment
Technical Expertise	High - understands system architecture
Primary Tasks	Tenant management, user impersonation, system configuration
Access Pattern	Web browser, secure workstation
Availability Need	Business hours with on-call for incidents

Key Permissions:

- Create/manage PSP tenants
- Impersonate any user
- Access all audit logs
- Configure system-wide settings
- Manage feature flags

PSP Admin

Attribute	Description
Description	PSP organization administrator managing brands and operations
Population	2-5 per PSP tenant

Technical Expertise	Medium - power user level
Primary Tasks	Brand onboarding, user management, reporting
Access Pattern	Desktop web browser
Availability Need	Business hours

Key Permissions:

- Create/manage brands
- Manage all PSP-level users
- Access all brands within tenant
- Generate reports and exports
- Configure PSP settings

Production Operator

Attribute	Description
Description	PSP staff managing fulfillment operations
Population	5-20 per PSP tenant
Technical Expertise	Low-Medium - task-focused
Primary Tasks	Order processing, shipment creation, batch management
Access Pattern	Desktop web browser, possibly warehouse terminal
Availability Need	Business hours, shift-based

Key Permissions:

- View/update order status
- Create shipments and tracking
- Manage production batches
- Process reorders
- Read-only campaign access

Brand Admin

Attribute	Description
Description	Brand's primary system administrator
Population	1-3 per brand
Technical Expertise	Medium - marketing/operations background
Primary Tasks	Store management, campaign creation, user permissions
Access Pattern	Desktop web browser
Availability Need	Business hours

Key Permissions:

- Full brand configuration
- Create/manage campaigns
- Manage brand users
- Access all stores
- Review and approve photos

Campaign Manager

Attribute	Description
Description	Brand staff focused on campaign execution
Population	2-10 per brand
Technical Expertise	Low-Medium - marketing background
Primary Tasks	Campaign setup, store assignment, progress monitoring
Access Pattern	Desktop web browser
Availability Need	Business hours

Key Permissions:

- Create/edit assigned campaigns
- Manage store assignments
- View campaign analytics
- Cannot manage brand settings
- Scoped to assigned campaigns only

Regional Manager

Attribute	Description
Description	Field manager overseeing store compliance
Population	5-20 per brand (varies by brand size)
Technical Expertise	Low - field operations background
Primary Tasks	Photo review, exception handling, store support
Access Pattern	Tablet or laptop, mobile-friendly web
Availability Need	Business hours, field schedule

Key Permissions:

- Review photos for assigned regions
- Approve/reject submissions
- Handle escalations
- Read-only store data
- Scoped to assigned regions

Store Manager

Attribute	Description
-----------	-------------

Description	Store-level authority for campaign execution
Population	1 per store (1,000+ per brand)
Technical Expertise	Low - retail operations background
Primary Tasks	Team coordination, approval authority, issue escalation
Access Pattern	Mobile device (phone/tablet), PWA
Availability Need	Store hours

Key Permissions:

- Full store execution access
- Approve replacement requests
- Manage store team
- Complete all surveys
- Report issues

Store Operator

Attribute	Description
Description	Store staff performing installation tasks
Population	1-5 per store
Technical Expertise	Low - basic smartphone proficiency
Primary Tasks	Receive shipments, install materials, capture photos
Access Pattern	Mobile device (phone), PWA
Availability Need	Store hours

Key Permissions:

- Execute surveys
- Upload photos
- Update task status
- Request replacements
- Cannot manage other users

Integration User

Attribute	Description
Description	Service account for system-to-system communication
Population	1-5 per PSP tenant
Technical Expertise	N/A - automated
Primary Tasks	API operations, webhook receipt, data sync
Access Pattern	REST API only

Availability Need	24/7 automated
-------------------	----------------

Key Permissions:

- API access per granted scopes
- No UI access
- Webhook endpoint access
- Rate-limited operations
- Audit-logged actions

User Class Interaction Matrix

From \ To	Platform	PSP	Brand	Regional	Store
Platform Admin	Manage	Manage	View	View	View
PSP Admin	-	Manage	Manage	View	View
Brand Admin	-	-	Manage	Manage	View
Regional Manager	-	-	-	Self	View
Store Manager	-	-	-	-	Manage

Accessibility Considerations

User Class	Accessibility Needs
Store Operators	Large touch targets, simple navigation, outdoor visibility
Regional Managers	Tablet-optimized, minimal typing
All Web Users	WCAG 2.1 AA compliance, keyboard navigation
All Users	Screen reader compatibility for core functions

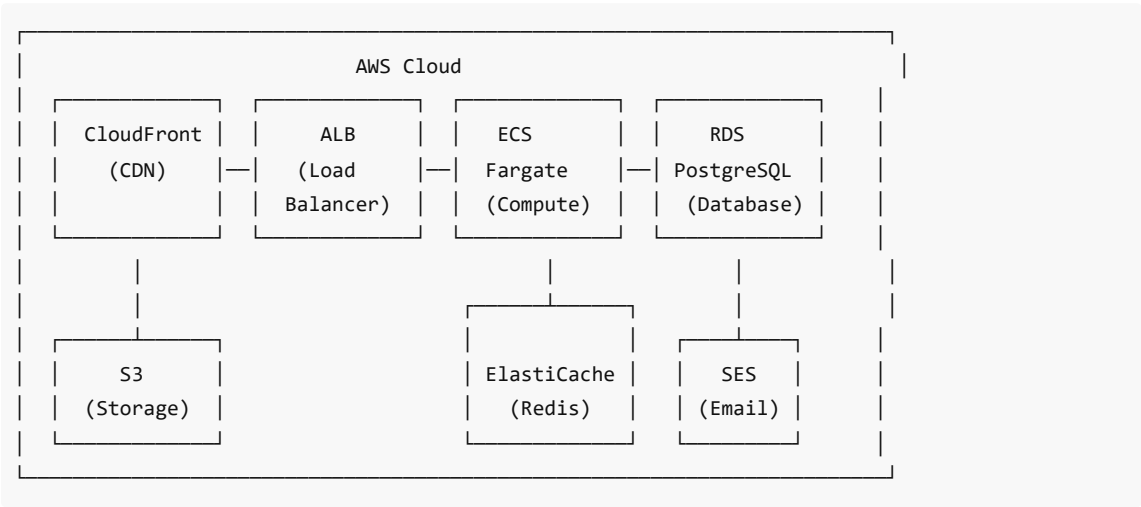
Training Requirements

User Class	Training Level	Delivery Method
Platform Admin	Comprehensive	Technical documentation, hands-on
PSP Admin	Full	Online training, documentation
Production Operator	Role-specific	Workflow guides, quick reference
Brand Admin	Full	Online training, webinar
Campaign Manager	Role-specific	Interactive tutorials
Regional Manager	Task-focused	Mobile-optimized guides
Store Manager	Task-focused	In-app guidance, video
Store Operator	Minimal	In-app walkthrough

2.4 Operating Environment

Infrastructure Overview

NewPOPSys v1.38 operates as a cloud-native application deployed on AWS infrastructure with a focus on reliability, scalability, and cost efficiency.



Production Environment Specifications

Compute Resources

Component	Service	Configuration
Web/API Servers	ECS Fargate	2 vCPU, 4GB RAM per task
Background Workers	ECS Fargate	2 vCPU, 4GB RAM per task
Minimum Tasks	Auto-scaling	2 tasks per service (HA)
Maximum Tasks	Auto-scaling	10 tasks per service

Database

Attribute	Specification
Engine	PostgreSQL 15+
Service	AWS RDS
Instance Class	db.r6g.large (minimum)
Storage	100GB gp3 SSD (expandable)

Multi-AZ	Yes (production)
Read Replicas	1 (optional, for reporting)
Backup Retention	7 days automated

Cache & Queue

Attribute	Specification
Engine	Redis 7+
Service	AWS ElastiCache
Node Type	cache.r6g.large
Cluster Mode	Disabled
Replication	1 replica (HA)
Purpose	Session cache, BullMQ jobs

Storage

Type	Service	Configuration
Photo Storage	S3 Standard	Versioning enabled
Export Artifacts	S3 Standard-IA	30-day lifecycle
Static Assets	S3 + CloudFront	CDN distribution
Backup Archives	S3 Glacier	Long-term retention

Capacity Targets

Pilot Scale (v1.38)

Dimension	Target	Notes
PSP Tenants	2	Visual Graphx, Speedy CPS
Brands per PSP	2-3	Good2Go confirmed
Stores per Brand	Up to 1,000	Primary scale factor
Concurrent Users	50	Peak during campaign launch
Photos per Campaign	1+ per item per slot	Variable by campaign
Data Retention	90 days	Post campaign completion

Performance Baselines

Operation	p50 Target	p95 Target	p99 Target
-----------	------------	------------	------------

Simple Read	<50ms	<150ms	<300ms
Complex Read	<100ms	<300ms	<600ms
Write Operations	<150ms	<400ms	<800ms
Photo Upload	<2s	<5s	<10s
Report Generation	<30s	<60s	<120s

Client Environment Requirements

Web Application (Desktop)

Requirement	Specification
Browsers	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
Screen Resolution	1280x720 minimum, 1920x1080 recommended
Network	Broadband internet (5+ Mbps)
JavaScript	Required, ES2020+ support
Cookies	Required for session management
Local Storage	Required for preferences

Mobile PWA (Store Execution)

Requirement	Specification
Operating System	iOS 14+, Android 10+
Browser	Safari (iOS), Chrome (Android)
Screen Size	5" minimum, touch-enabled
Camera	8MP minimum, autofocus
Storage	500MB available for offline cache
Network	4G LTE or WiFi (sync-on-open)

Network Requirements

Protocol	Port	Purpose
HTTPS	443	All application traffic
WSS	443	WebSocket notifications

Firewall/Proxy Requirements:

- No SSL inspection (breaks certificate pinning)
- WebSocket upgrade support

- No request size limits <10MB

Software Dependencies

Server-Side Stack

Component	Version	Purpose
Node.js	20 LTS	Runtime
TypeScript	5.x	Language
Next.js	14.x	Web framework
Fastify	4.x	API server
Drizzle ORM	Latest	Database ORM
BullMQ	4.x	Job queue
Zod	3.x	Validation

Client-Side Stack

Component	Version	Purpose
React	18.x	UI framework
TanStack Query	5.x	Data fetching
Tailwind CSS	3.x	Styling
Radix UI	Latest	Accessible components

Observability Stack

Component	Purpose
OpenTelemetry	Distributed tracing
Sentry	Error tracking
AWS CloudWatch	Metrics and logs

Security Environment

Network Security

Control	Implementation
TLS Version	1.3 only
Certificate	ACM-managed
WAF	AWS WAF with OWASP rules

DDoS Protection	AWS Shield Standard
-----------------	---------------------

Data Security

Data State	Protection
In Transit	TLS 1.3
At Rest (DB)	AES-256 (RDS encryption)
At Rest (S3)	AES-256 (SSE-S3)
At Rest (Redis)	AES-256 (ElastiCache encryption)

Authentication

Attribute	Specification
Token Type	JWT (RS256)
Access Token TTL	15 minutes
Refresh Token TTL	7 days
Password Policy	12+ chars, complexity required
MFA	Optional (v1), platform admin required

Availability Requirements

Metric	Target	Measurement
Uptime	99.5%	Monthly
Allowed Downtime	3.6 hours/month	Calculated
RTO (Single AZ)	<5 minutes	Recovery time
RPO	1 hour	Data loss tolerance

Maintenance Windows

Window	Timing	Impact
Planned Maintenance	Sunday 02:00-06:00 UTC	Potential brief outages
Emergency Patches	As needed	Rolling deployment
Database Maintenance	RDS automated	Usually zero downtime

Environment Tiers

Environment	Purpose	Scale
-------------	---------	-------

Development	Local development	Single instance
Staging	Integration testing	50% production
Production	Live operations	Full scale

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2.5 Design and Implementation Constraints

Architectural Constraints

Multi-Tenancy Model

Constraint	Rationale
PSP as root tenant	Business model positions PSP as paying customer
Logical data isolation	Single database with tenant_id column pattern
No cross-tenant data access	Compliance and data privacy requirements
Brand scoping within tenant	Hierarchical access control

Technology Stack Mandates

Component	Mandated Choice	Rationale
Runtime	Node.js 20 LTS	Team expertise, ecosystem
Language	TypeScript 5.x	Type safety, maintainability
Web Framework	Next.js 14.x	Full-stack React, SSR support
API Framework	Fastify 4.x	Performance, schema validation
Database	PostgreSQL 15+	JSONB, advanced features
ORM	Drizzle	Type-safe queries, performance
Queue	BullMQ/Redis	Proven reliability
Cloud	AWS	Existing infrastructure

API Design Constraints

Constraint	Specification
Protocol	REST over HTTPS
Authentication	JWT Bearer tokens
Versioning	URL path versioning (/v1/)

Content Type	application/json
Idempotency	Required for all write operations
Rate Limiting	Per-tenant, per-endpoint

Regulatory and Compliance Constraints

Accessibility Requirements

Standard	Requirement	Impact
WCAG 2.1 AA	All web interfaces	UI component selection
Section 508	Federal accessibility	Government client support
Keyboard Navigation	Full functionality	No mouse-only features
Screen Reader	Core workflows	Semantic HTML required

Data Privacy

Regulation	Applicability	Constraint
Data Retention	90-day post-completion	Automated purge required
Right to Export	Before deletion	Export-before-delete workflow
Audit Logging	All data modifications	Immutable audit trail
PII Handling	User data, store contacts	Encryption at rest

Security Standards

Standard	Application
OWASP Top 10	Security testing baseline
TLS 1.3	All external connections
Password Policy	NIST 800-63B aligned
Session Management	Secure token handling

Operational Constraints

Deployment Constraints

Constraint	Specification
Zero-Downtime Deploy	Rolling deployments required
Blue-Green Capability	Database migrations must be backward-compatible

Feature Flags	New features behind flags
Rollback Window	15-minute rollback capability

Performance Constraints

Constraint	Target
API Response (p95)	<150ms for simple reads
API Response (p95)	<300ms for complex reads
Photo Upload	<5s for 10MB image
Page Load	<3s initial, <1s subsequent
Offline Sync	<30s for typical session

Scalability Constraints

Dimension	Pilot Constraint	Future Capacity
Stores per Brand	1,000	10,000
Concurrent Users	50	500
Photos per Campaign	10,000	100,000
API Requests	100 req/s	1,000 req/s

Business Constraints

Scope Boundaries (Protected Not-Now)

Category	Constraint	Rationale
Not MIS/ERP	No costing, scheduling, accounting	Integration point only
Not Marketplace	No bidding, matching, ratings	Single PSP model
Not Analytics	No predictive/AI insights	Data readiness only
Not Archive	No storage beyond 90 days	Export pathways provided
Not Customizable	Limited store UX flexibility	PSP adoption priority

Pilot Constraints

Constraint	Value
Pilot PSPs	2 (Visual Graphx, Speedy CPS)
Pilot Duration	90-day evaluation period
Scope Changes	Formal approval required

Feature Additions	Deferred to post-pilot
-------------------	------------------------

Integration Constraints

External System Integration

System	Constraint
PSP MIS	Async only (eventual consistency)
Shipping Carriers	Tracking data normalization required
Email Delivery	AWS SES limits apply
Webhook Endpoints	Customer-provided, HTTPS only

API Compatibility

Constraint	Specification
Webhook Signing	HMAC-SHA256 required
Retry Policy	Exponential backoff, 24h max
Payload Size	1MB maximum
Timeout	30-second response required

User Interface Constraints

Browser Support

Browser	Minimum Version	Support Level
Chrome	90+	Full
Firefox	88+	Full
Safari	14+	Full
Edge	90+	Full
IE11	Not supported	None

Mobile Constraints

Platform	Constraint
iOS	Safari only (WebKit)
Android	Chrome primary, WebView fallback
Offline	Sync-on-open, draft storage only

Camera	Native camera access required
--------	-------------------------------

Responsive Design

Breakpoint	Target Devices
<640px	Mobile phones
640-1024px	Tablets
> 1024px	Desktop

Development Constraints

Code Quality

Constraint	Requirement
Test Coverage	80% line coverage
Linting	ESLint with strict config
Type Safety	No implicit any
Code Review	Required for all PRs

Documentation

Artifact	Requirement
API Docs	OpenAPI 3.0 specification
Code Comments	JSDoc for public interfaces
ADRs	Architecture decisions documented
Runbooks	Operational procedures

Version Control

Constraint	Specification
SCM	Git
Branching	Trunk-based with feature flags
Commit Messages	Conventional commits
Release Tags	Semantic versioning

Resource Constraints

Team Expertise

Skill Area	Available	Constraint
TypeScript/Node.js	Strong	Primary stack
React/Next.js	Strong	UI framework
PostgreSQL	Strong	Database
AWS	Medium	Infrastructure
Mobile/PWA	Medium	Store execution

Timeline Constraints

Milestone	Constraint
MVP Delivery	Fixed date
Feature Freeze	2 weeks before pilot
Bug Fix Window	Pilot duration
Post-Pilot Roadmap	Based on feedback

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2.6 Assumptions and Dependencies

Assumptions

Business Assumptions

ID	Assumption	Impact if Invalid
BA-01	PSP is the paying customer and orchestrating authority	Business model restructure required
BA-02	Brands operate within PSP tenant context	Multi-PSP brand support needed
BA-03	Stores have basic smartphone access for execution	Alternative input methods needed
BA-04	Campaign cycles are predictable (seasonal, promotional)	Capacity planning adjustments
BA-05	Photo proof is acceptable verification evidence	Alternative verification mechanisms
BA-06	90-day data retention is acceptable post-completion	Extended retention architecture
BA-07	Pilot brands will provide timely feedback	Pilot timeline extension

Technical Assumptions

ID	Assumption	Impact if Invalid
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TA-01	AWS services remain available in target regions	Multi-cloud architecture
TA-02	PostgreSQL 15+ features are sufficient	Database migration
TA-03	Node.js 20 LTS supported through pilot	Runtime upgrade
TA-04	Mobile browsers support PWA adequately	Native app development
TA-05	Redis provides sufficient queue reliability	Alternative queue system
TA-06	S3 provides adequate photo storage performance	CDN optimization
TA-07	JWT tokens are acceptable for authentication	Session-based auth

User Assumptions

ID	Assumption	Impact if Invalid
UA-01	Store operators have basic digital literacy	Enhanced training, simplified UX
UA-02	Regional managers can perform photo review	PSP-level review workflow
UA-03	Brand admins understand campaign concepts	Extensive onboarding
UA-04	PSP staff can manage production workflows	Workflow simplification
UA-05	Users have reliable email access	Alternative notification channels
UA-06	Mobile devices have functional cameras	Barcode-only verification

Operational Assumptions

ID	Assumption	Impact if Invalid
OA-01	Internet connectivity available at stores	Enhanced offline capability
OA-02	Stores can receive shipments reliably	Alternative delivery workflows
OA-03	PSP can provide tracking information	Manual status updates
OA-04	Campaigns have defined start/end dates	Indefinite campaign support
OA-05	Issue reporting happens within campaign window	Extended issue window
OA-06	Photo upload happens same-day as installation	Extended upload windows

Dependencies

External Service Dependencies

ID	Dependency	Provider	Criticality	Fallback
ED-01	Compute Infrastructure	AWS ECS Fargate	Critical	None (single cloud)
ED-02	Database Service	AWS RDS PostgreSQL	Critical	Manual restore from backup

ED-03	Cache/Queue Service	AWS ElastiCache Redis	High	Degraded mode (no cache)
ED-04	Object Storage	AWS S3	Critical	None
ED-05	CDN	AWS CloudFront	Medium	Direct S3 access
ED-06	Email Delivery	AWS SES	Medium	Alternative SMTP
ED-07	DNS	AWS Route 53	Critical	None
ED-08	SSL Certificates	AWS ACM	Critical	None

Third-Party Integration Dependencies

ID	Dependency	Purpose	Criticality	Fallback
TI-01	PSP MIS Systems	Order/status sync	Medium	Manual data entry
TI-02	Shipping Carriers	Tracking information	Low	Manual tracking entry
TI-03	Brand Systems	Store data sync	Low	CSV import
TI-04	Sentry	Error tracking	Low	CloudWatch only
TI-05	OpenTelemetry	Distributed tracing	Low	Local logging only

Development Dependencies

ID	Dependency	Version	Purpose	Risk
DD-01	Node.js	20 LTS	Runtime	Low (LTS)
DD-02	TypeScript	5.x	Language	Low
DD-03	Next.js	14.x	Framework	Medium (major version)
DD-04	React	18.x	UI Library	Low
DD-05	Drizzle ORM	Latest	Database access	Medium (active development)
DD-06	BullMQ	4.x	Job queue	Low
DD-07	Zod	3.x	Validation	Low
DD-08	Tailwind CSS	3.x	Styling	Low

Infrastructure Dependencies

ID	Dependency	Purpose	SLA Required
ID-01	AWS Region Availability	All services	99.99%
ID-02	RDS Multi-AZ	Database HA	99.95%
ID-03	ElastiCache Replication	Cache HA	99.9%
ID-04	S3 Durability	Data storage	99.999999999%

ID-05	CloudFront Edge	Asset delivery	99.9%
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Dependency Risk Assessment

Critical Path Dependencies

```
graph TD
    A[User Request] --> B[CloudFront/ALB]
    B --> C[ECS Fargate]
    C --> D[RDS PostgreSQL]
    C --> E[ElastiCache Redis]
    C --> F[S3 Storage]

    style D fill:#ff6666
    style C fill:#ff6666
    style B fill:#ff6666

    subgraph "Critical Path"
        B
        C
        D
    end

    subgraph "High Availability"
        E
        F
    end

    end
```

Dependency Monitoring

Dependency	Monitoring Method	Alert Threshold
RDS	CloudWatch	CPU >80%, Connections >80%
ECS	CloudWatch	Task failures, Memory >85%
ElastiCache	CloudWatch	Memory >75%, Connections >80%
S3	CloudWatch	5xx errors >1%
External APIs	Health checks	Response time >5s

Contingency Planning

Dependency Failure Scenarios

Scenario	Impact	Mitigation
RDS Primary Failure	Service outage	Multi-AZ automatic failover
ElastiCache Failure	Degraded performance	Cache bypass, direct DB

S3 Unavailable	Photo upload fails	Queue uploads, retry
SES Throttling	Email delays	Queue emails, batch send
Single AZ Failure	Potential outage	Multi-AZ deployment

Recovery Procedures

Scenario	RTO	RPO	Procedure
Database Failure	<5 min	0 (sync replica)	Automatic failover
Application Failure	<2 min	N/A	ECS task restart
Region Failure	<4 hours	1 hour	Cross-region restore
Data Corruption	<1 hour	Point-in-time	PITR restore

Pilot-Specific Dependencies

Pilot Partner Dependencies

Partner	Dependency	Requirement
Visual Graphx	MIS integration specs	API documentation
Speedy CPS	System access	Staging credentials
Good2Go (Brand)	Store data	Complete store list
All Pilot Partners	Feedback sessions	Weekly availability

Pilot Success Dependencies

ID	Dependency	Owner	Due Date
PS-01	Store list finalized	Brand Admin	Pilot -14 days
PS-02	Survey templates configured	Brand Admin	Pilot -7 days
PS-03	User accounts provisioned	PSP Admin	Pilot -3 days
PS-04	Training completed	All users	Pilot -1 day
PS-05	Integration testing complete	Dev team	Pilot -7 days